

WT - Exp – 3

Soham Satpute - 51

Prajyot Shinde- 55

Laksh Sodhai - 57

Div: D15A

Project name: Radar Detection System.

Aim:

To design and implement a radar system using Arduino and ultrasonic sensors to detect objects within a specified range and display real-time results for educational and experimental purposes.

Introduction:

Radar (Radio Detection and Ranging) systems are widely used for detecting, tracking, and measuring the speed of objects. They work by emitting radio waves and analyzing the reflected signals to determine an object's distance, position, and movement. Traditional radar systems use electromagnetic waves, but in this project, we will simulate a radar system using an ultrasonic sensor mounted on a servo motor. The ultrasonic sensor emits sound waves, and based on the time taken for the waves to bounce back from an obstacle, the distance and position of the object can be determined.

This project leverages the Arduino microcontroller to control the radar system and display real-time results. By using a servo motor, we can scan a wide area and detect objects in different directions. The collected data is processed and visualized on a computer screen in a radar-like interface, making it an excellent learning tool for students and hobbyists.

Objectives:

- To design a cost-effective and simple radar system using Arduino and ultrasonic sensors.
- To detect and display object positions in real-time using sensor data.
- To implement a scanning mechanism using a servo motor for a wider field of detection.
- To develop a graphical interface that simulates a traditional radar screen.
- To enhance practical knowledge of embedded systems, Arduino programming, and sensor interfacing.
- To explore applications of radar-like systems in security, automation, and robotics.

Challenges:

- Limited detection range and accuracy: The HC-SR04 ultrasonic sensor has a maximum range of about 4 meters and may have difficulties detecting small or distant objects.
- Environmental Interference: The accuracy of the sensor can be affected by temperature, humidity, and obstacles with irregular surfaces.
- Real-time Data Processing: Processing and displaying the data in a radar-like graphical interface without delay requires efficient coding and hardware optimization.

- **Angular Precision:** The servo motor's resolution determines the accuracy of the scanning mechanism, which can be a limiting factor in object detection.
 - **Hardware Constraints:** The Arduino board has limited processing power and memory, which can restrict advanced functionalities.
-

Solution Approach:

1. **Component Selection:** The system consists of an Arduino board, an HC-SR04 ultrasonic sensor for distance measurement, and an SG90 servo motor for scanning.
2. These components are chosen for their affordability and ease of integration.
3. **Hardware Setup:** The ultrasonic sensor is mounted on a servo motor, allowing it to rotate and scan the surroundings. The setup is connected to the Arduino, which processes the data and controls the servo's movement.
4. **Programming:** An Arduino script is developed to control the servo motor, capture sensor readings, and process the data. Serial communication is used to send the processed information to a connected computer for display.
5. **Data Processing & Visualization:** A graphical user interface (GUI) is created using Processing or Python to display detected objects in a radar-like format. The interface updates in real time as the servo scans the area.
6. **Testing & Calibration:** The system is tested to fine-tune the scanning speed, sensor accuracy, and data visualization. Adjustments are made to optimize performance and minimize detection errors.

Scope:

- **Educational & Research Applications:** This project serves as an effective learning tool for students and researchers to understand the working of radar systems and sensor-based detection mechanisms.
- **Automation & Security Systems:** The radar system can be adapted for motion detection, obstacle avoidance, and surveillance applications in industries and home automation.
- **Integration with Advanced Technologies:** The system can be expanded by integrating IoT, AI, or machine learning to improve detection accuracy and functionality.

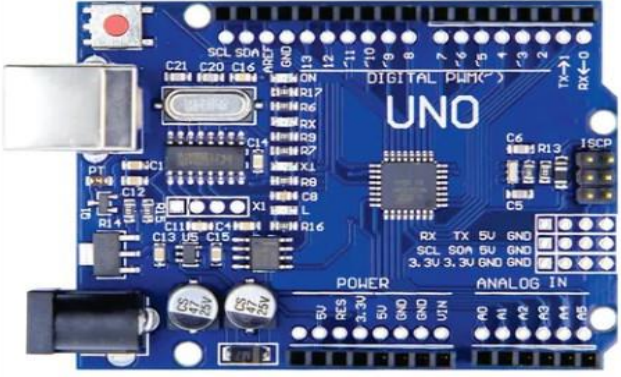
Advantages:

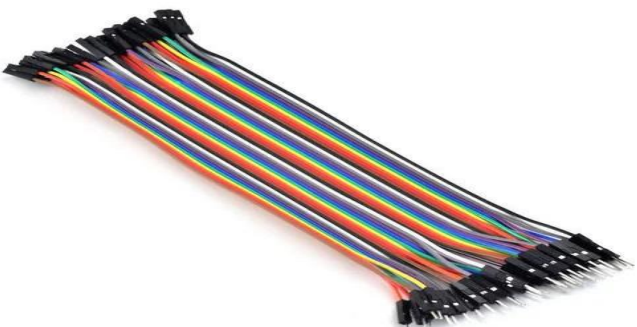
- **Cost-Effective:** Uses inexpensive components, making it accessible for students and hobbyists.
- **Educational Value:** Helps in understanding radar principles, sensor interfacing, and Arduino programming.
- **Real-Time Detection:** Provides immediate feedback and visualization of objects within range.
- **Customizable & Expandable:** Can be enhanced by integrating additional sensors or microcontrollers.
- **Useful for Security & Automation:** Can be used for motion detection, intrusion alerts, and obstacle avoidance in robotics.
- **Low Power Consumption:** Operates efficiently using a microcontroller and low-power.

Conclusion:

The Arduino-based radar system is an efficient and cost-effective solution for object detection and monitoring. It provides hands-on experience with embedded systems, sensor integration, and real-time data visualization. Despite some limitations in range and accuracy, it is a valuable educational tool with potential applications in security and automation. Future enhancements could include IoT connectivity and AI integration to improve performance. This project serves as a foundation for exploring more advanced radar technologies.

Components:

Sr no.	Description	Image
1.	Arduino Uno SMD A microcontroller board based on the ATmega328P, designed for embedded applications with digital and analog I/O pins. It processes data from sensors and controls the servo motor to classify objects as wet or dry.	 The image shows an Arduino Uno SMD (Surface Mount Device) board. It is a blue printed circuit board (PCB) with various components including a USB Type-C port, a DC power jack, a reset button, and numerous integrated circuits and passive components. The board is labeled 'UNO' and 'DIGITAL POWER'.
2.	Servo 9g Motor A small, lightweight servo motor capable of precise angular movement, typically used in automation and robotics. In this system, it rotates left or right to direct the object to its designated category.	 The image shows a small, blue plastic servo motor with a black label that reads 'MG90S 9g'. It has a red, white, and brown three-wire cable. Next to the servo are several white plastic gears of different sizes and a small metal screw.
3.	Ultrasonic Sensor A distance-measuring sensor that emits high-frequency sound waves and calculates the time delay of reflected waves. It detects the presence of an object, triggering the classification process.	 The image shows an ultrasonic sensor module mounted on a blue PCB. It features two large, cylindrical metal horns for emitting and receiving sound waves. The module has three pins at the bottom and is labeled 'SR04'.

5.	USB Cable A standard cable used to supply power and enable data communication between the Arduino and a computer. It facilitates programming and real-time monitoring of the system.	
6.	M-F Jumpers Flexible connectors with male-to-female terminals that establish electrical connections between the Arduino, sensors, and motor. They ensure proper signal transmission and circuit functionality.	
7.	Breadboard A solderless prototyping board used for building and testing electronic circuits. It enables easy connections between the Arduino, sensors, and motor without permanent wiring.	